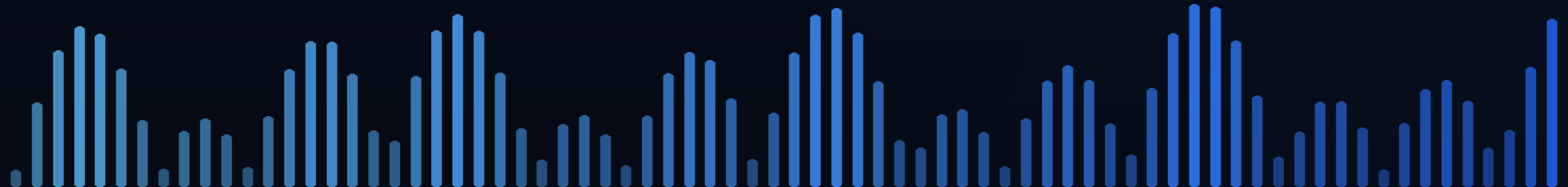


Voice Signal Analysis Using Machine Learning for Early Detection of Parkinson's Disease

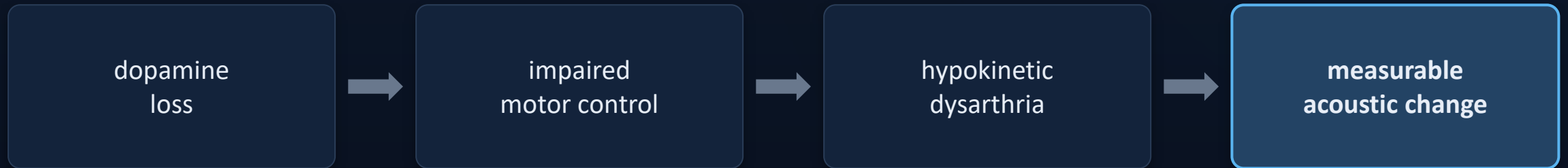
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ID 2200208522 Supervisor: Dr. Adel Adyaf

Control and Instrumentation Division • Department of Biomedical Engineering
Faculty of Engineering, University of Tripoli • Spring 2026



A movement disease reaches the voice



every muscle — breathing, larynx, articulation

The weakness that shakes a hand also moves the voice.

The voice is a cheap, early window



~90%

of patients show
speech changes



5 yr

before diagnosis,
in one documented case

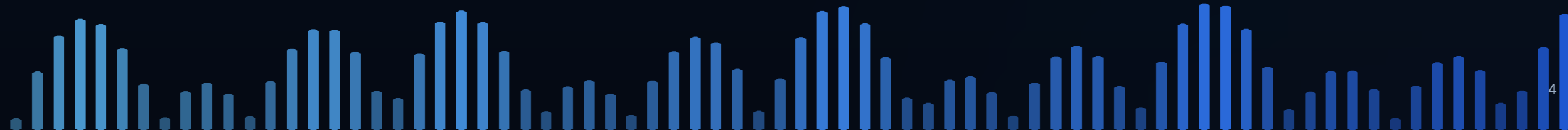


\$

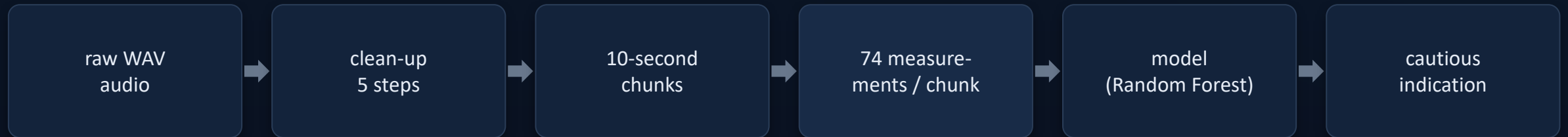
a microphone:
non-invasive, repeatable

Not a diagnosis — a reason to see a doctor earlier.

Can a computer measure a voice recording
and tell whether its pattern resembles
Parkinson's patients — or healthy speakers?



Six stages, one shared implementation



*The same code runs in training, testing, and prediction — **enforced, not promised.***

Every measurement points at physiology



monotone speech → **pitch variation**

unstable vocal-fold control → **jitter, shimmer**

incomplete glottal closure → **HNR, CPPS**

disturbed timing → **pause statistics**

imprecise articulation → **MFCC dynamics**

*All 74 measurements implemented in this project — **none downloaded.***

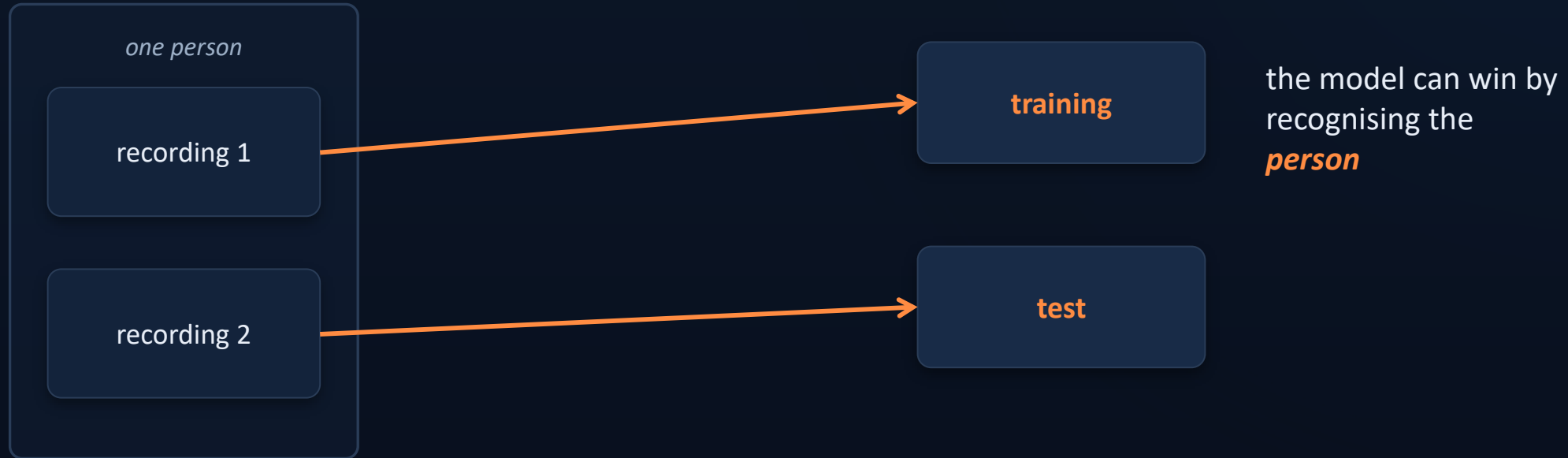
We chose *not* to clean the sound



Denoising suppresses irregularity.
Irregularity *is* the evidence.

Cleaning would make a disordered voice look artificially healthy.

A student with the exam questions proves nothing



This is **data leakage**. Our program halts if it ever happens.

Validation design decides the number

0.909

balanced accuracy

an impressive result?

Validation design decides the number

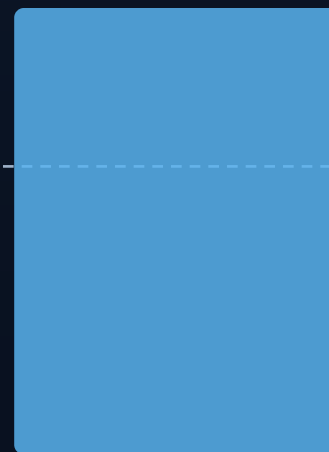
0.909

deliberately wrong split — demonstration only

0.775

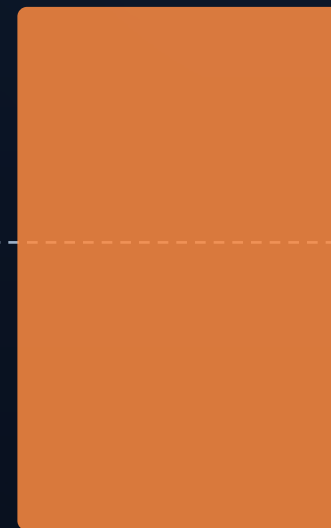
correct split, same model

0.775



correct split
grouped by subject

0.909



wrong split
chunks cross sides

The difference is **memorisation**, not detection — and it is why 95%+ results exist.

The rules were fixed before the experiments



identical splits for every variant



adoption margin: +0.02



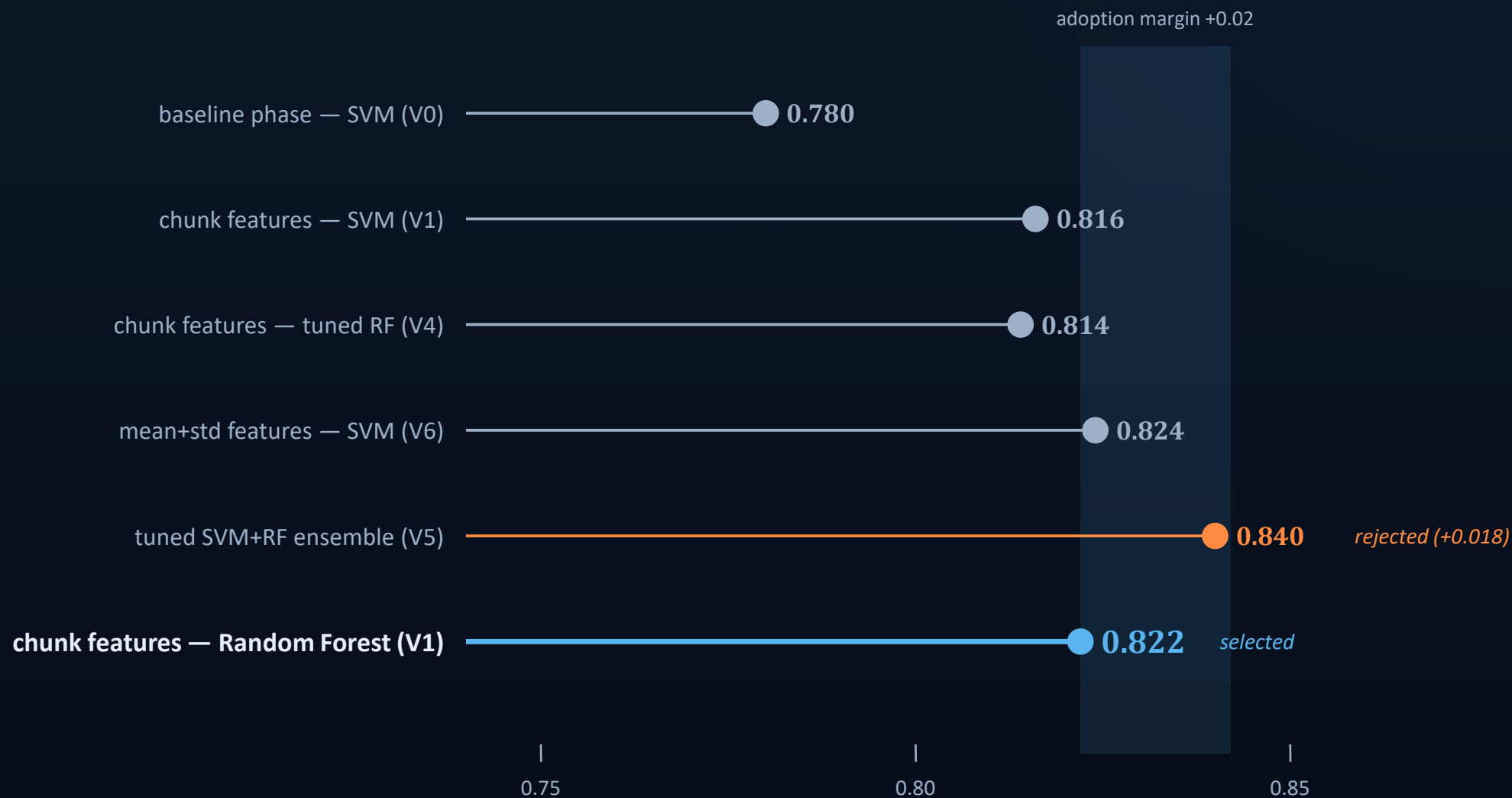
any result ≥ 0.95 : stop, investigate



all 19 configurations reported

Best score **0.840** — **rejected**. It won by 0.018: less than the noise.

Nineteen configurations — the winner chosen by rule



subject-level balanced accuracy, 3-seed mean — six of the 19 configurations shown; all are reported

Judged on people it has never heard

raw audio

›

clean-up

›

10-s chunks

›

74 measures

›

model

›

indication

0.822 ± 0.032

balanced accuracy, per subject

0.771

sensitivity

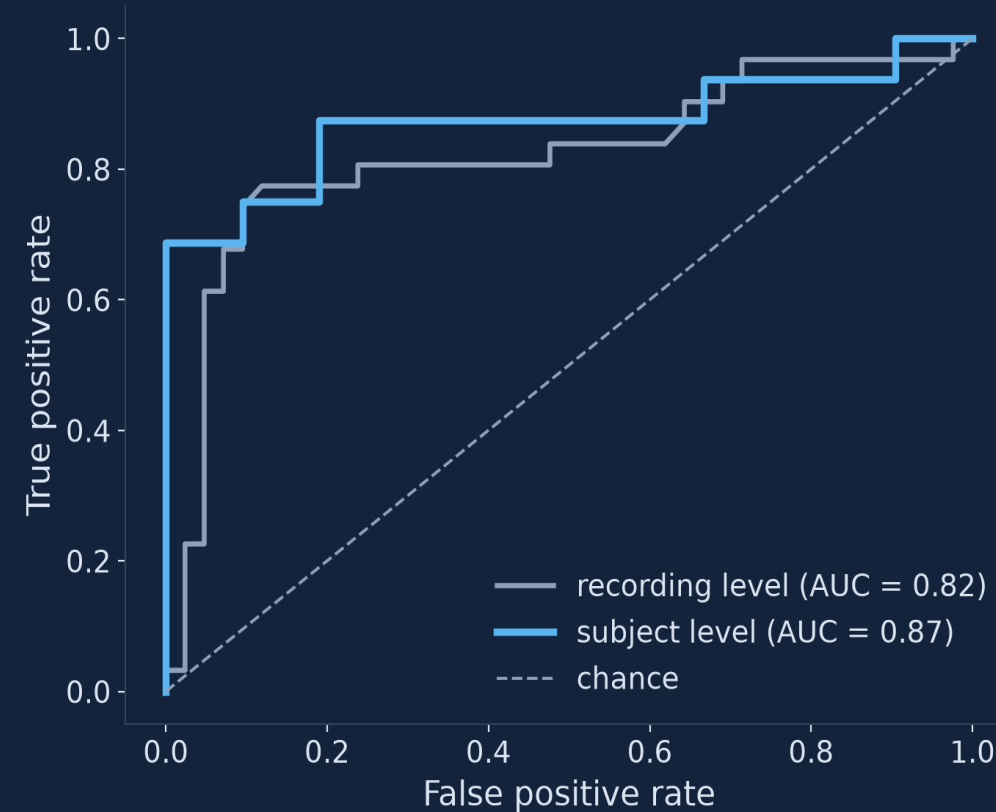
0.873

specificity

0.864

AUC

“always healthy” scores 0.575 accuracy, finds nobody — balanced accuracy 0.500.



It rediscovered a known clinical sign

raw audio

›

clean-up

›

10-s chunks

›

74 measures

›

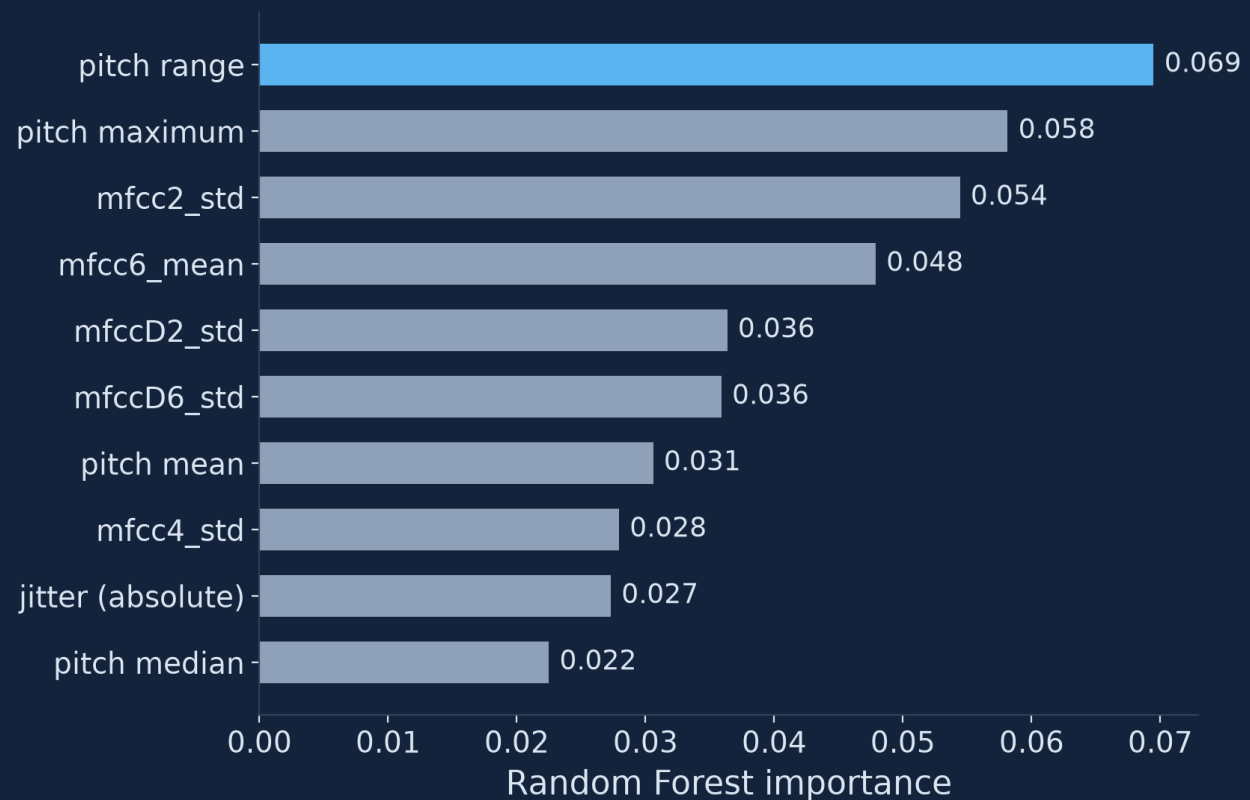
model

›

indication

Top measurement:
pitch range

the acoustic correlate of monotone speech —
described by clinicians for decades.



Italy: a frozen model, a trap, and a puzzle

raw audio

›

clean-up

›

10-s chunks

›

74 measures

›

model

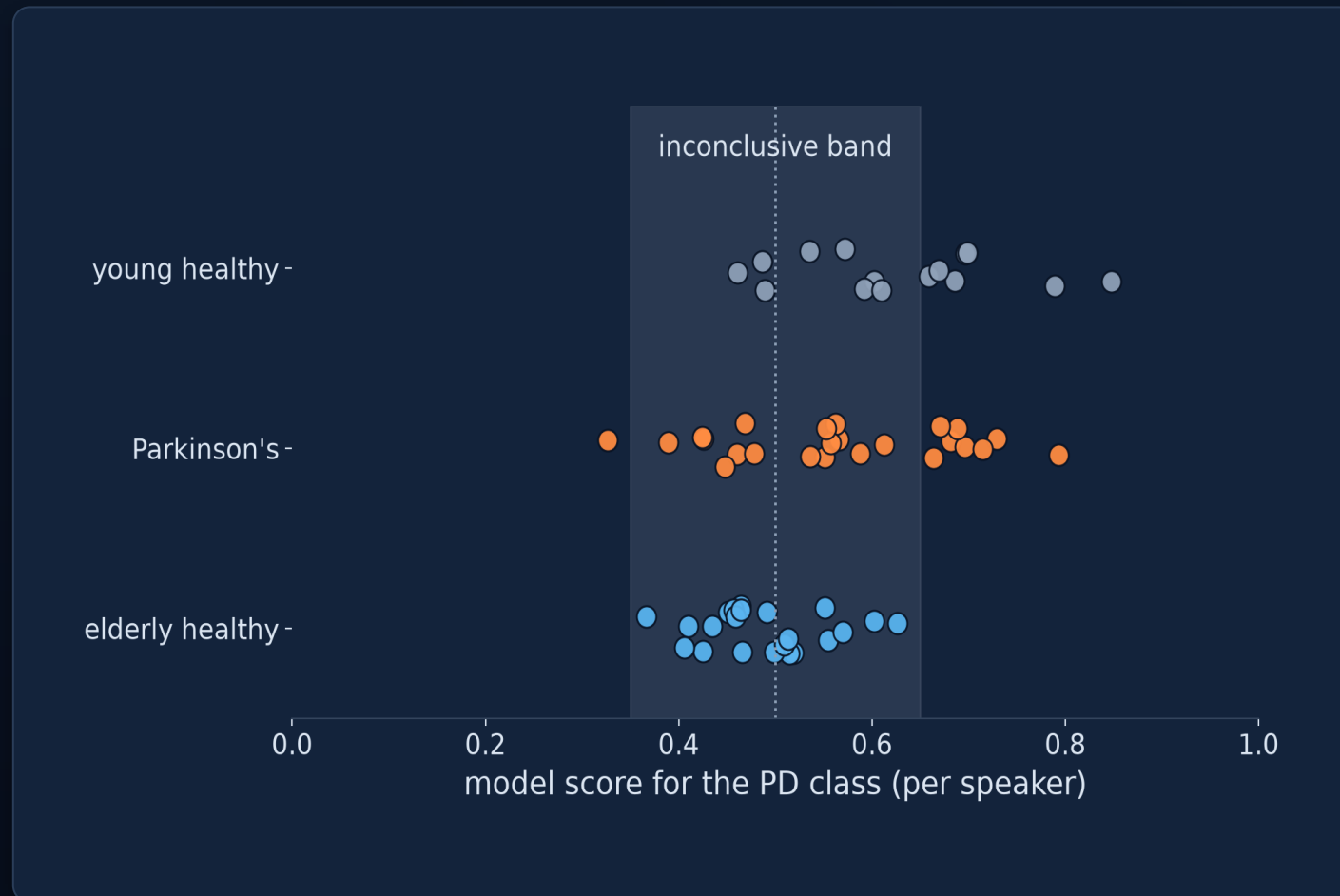
›

indication

Trap caught: every 44.1 kHz file was a patient.
Band-limited all audio to 16 kHz.

Result: AUC 0.701, balanced accuracy 0.629.

Puzzle: healthy young 0.626 > patients 0.566.
Why?



A system that refuses to guess

ID07_pd_2_0_0.wav
18.6MB

File information

- **File:** ID07_pd_2_0_0.wav
- **Duration:** 147.7 s (analyzed after silence trimming: 145.3 s)
- **Original sample rate:** 44100 Hz

Research model result

The acoustic pattern was classified by the research model as closer to the PD class.

Model score for the PD class: **0.96** (0 = closer to HC class, 1 = closer to PD class; scores between 0.35 and 0.65 are reported as inconclusive). This is a property of the research model, **not** a person's medical risk or probability of disease.

This is a research screening-support prototype and is not a medical diagnostic tool. Its result must not replace evaluation by a qualified healthcare professional.

- scores 0.35–0.65 $\Rightarrow$ “inconclusive”
- **100% of elderly healthy Italian speakers** landed there
- declining to answer is a **feature**

Optional: live demonstration

```
$ python scripts/predict_file.py ID07_pd_2_0_0.wav
```

RESEARCH MODEL RESULT (non-diagnostic)

Duration: 147.7 s, original sample rate 44100 Hz

The acoustic pattern was classified by the research model as closer to the PD class.

Model score for the PD class: 0.96

- safest demo: a corpus recording, command line
- analysis takes about one minute
- fallback: the screenshots on the previous slide carry the same content

skip this slide if time is short

The limits, stated plainly



37 subjects — one device, one language

the dominant limitation; every result is reported with its spread



vowel-defined measures on continuous speech

jitter, shimmer and HNR are noisier here; CPPS compensates only partly



scores are not calibrated probabilities

0.7 does not mean a 70% chance of disease — the wording enforces this



one external corpus — one data point

0.701 AUC measures generalisation once; it does not characterise it

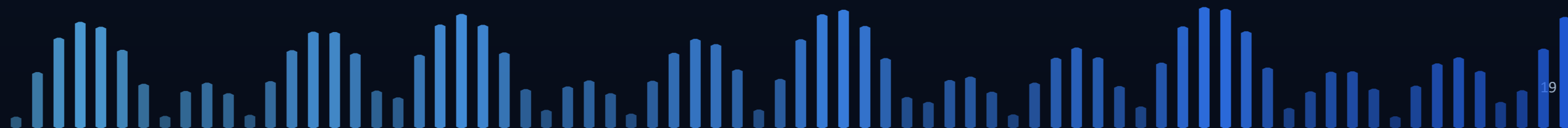
Every limitation above is stated in the report itself — none is left for an examiner to discover.

What this is — and is not

- not a diagnostic tool; many conditions change a voice
- 37 subjects, one language, one device

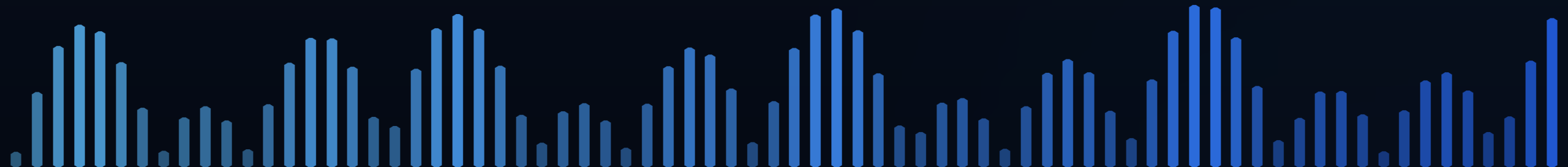
**Validation design moves the number
more than model engineering does.**

*An honest **0.822** is worth more than an inflated one.*



Thank you

Questions welcome.



Backup **Why 82% and not 95%?**

- published 95%+ figures usually split recordings or windows, not people
- our own pipeline: 0.909 leaky vs 0.775 honest — same model
- 0.822 is measured on people the model never heard

Backup **Why not deep learning?**

- 37 subjects is far below what deep networks need
- examiners can ask this model which measurements it used
- our small neural network collapsed in the confounder test (0.671 $\rightarrow$ 0.526)

Backup **Why balanced accuracy?**

- classes are imbalanced: 42 HC vs 31 PD recordings
- “always healthy” scores 0.575 accuracy and finds zero patients
- balanced accuracy gives that model 0.500 — correctly exposing it

Backup **Why reject the 0.840 ensemble?**

- the +0.02 adoption margin was fixed before the experiments
- 0.018 is smaller than the seed-to-seed noise (~ 0.03): a statistical tie
- we report 0.840 openly — we just did not let it drive the decision

Backup **Can it diagnose Parkinson's?**

- no — and it is not designed to
- colds, laryngitis, ageing, smoking also change the voice
- a non-diagnostic disclaimer appears before and after every result

Backup **Is 37 subjects too few?**

- yes — it is the main limitation, stated in every report
- we report variability, repeat with 3 seeds, use grouped validation
- and we tested externally on a second corpus of 61 speakers

Backup **Is it just detecting male vs female voices?**

- we tested it: removed all absolute-pitch features and re-ran everything
- the selected model held: 0.768 → 0.780
- the small neural network collapsed — and was excluded

Backup **Hardest technical problem?**

- in the Italian corpus, every 44.1 kHz file belonged to the patient group
- a model could “detect Parkinson's” by detecting microphone bandwidth
- caught at inspection; all audio band-limited to 16 kHz before evaluation